

NeckErgonomics: Adapting Head Rotation in HMDs with Predicted Neck Muscle Activity

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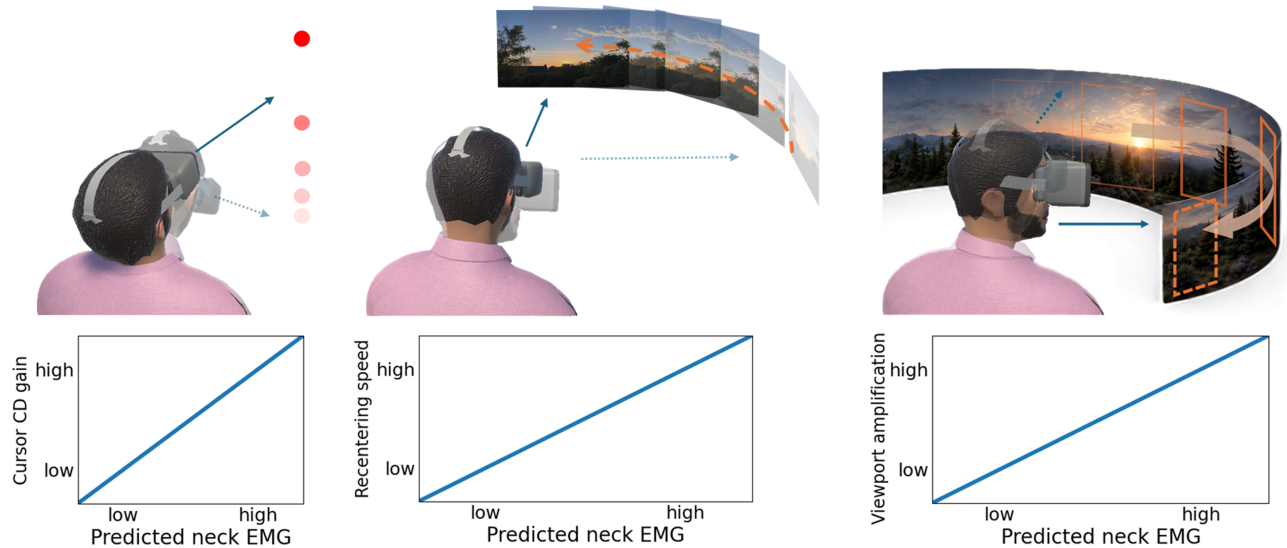


Figure 1: Three applications illustrate the concept of *NeckErgonomics* that adapts HMD interaction based on predicted neck electromyography (EMG) in the bottom line charts, depending on head pose and pose holding time. Left: Predicted neck EMG modulates the control–display gain of head cursor to amplify its movement during head extension. Centre: Predicted neck EMG modulates the recentering speed of a window during sustained attention, depending on holding time and head poses. Right: Predicted neck EMG modulates the viewport amplification to reduce required head rotation in yaw.

Abstract

Interaction with head-mounted displays (HMDs) places a significant workload on neck muscles, exacerbated by the extra weight of the HMD. This paper introduces *NeckErgonomics*, an interaction approach that adapts HMD interfaces according to estimated neck muscle electromyography (EMG) signals to improve neck

ergonomics. Leveraging an open-source dataset, we train a light-weight neural regression model to predict relative neck muscle activity from head yaw, head pitch, and fixation duration. We demonstrate *NeckErgonomics* through three model-informed applications: (1) control–display gain for head-based pointing, (2) dynamic recentering of virtual windows during knowledge work, and (3) adaptive viewport amplification for seated 360° video viewing. We evaluated the applications in a controlled within-subjects user study (N=18) to demonstrate the viability of *NeckErgonomics*. Across the three tasks, *NeckErgonomics* led to lower accumulated head movement, sustained head rotation, and perceived physical demand. The results also reveal task-specific trade-offs with speed, precision, visual stability, and sickness. Taken together, this work indicates a path toward ergonomics-by-design XR techniques in which adaptation



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is grounded in the estimated muscle activity rather than ambiguous behavioural cues.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Virtual reality**.

Keywords

Ergonomics, Head-Mounted Displays, Virtual Reality, Human Computer-Interaction, Neck Muscle Activity, Adaptive Interaction

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1 Introduction

Head rotation is central to the interaction model of head-mounted displays (HMDs). Interaction in HMDs induces a significant workload on the neck muscles to move and stabilise the head, including the additional weight of the HMD [4, 10, 31]. As Extended Reality (XR) moves beyond short demos toward productivity, training, and everyday use, reducing musculoskeletal strain becomes not only a comfort issue but also a usability and accessibility concern.

A promising way to improve neck ergonomics is adaptive XR interaction, in which interface parameters (e.g., cursor gain, display position) are dynamically adjusted to reduce effort. Prior work in HMDs improves neck ergonomics by dynamically repositioning content toward the user's forward viewing region [24, 26] and by dynamically modulating head-motion gain to reduce required head movement [9, 16, 18]. However, these adaptations typically infer physical demand from geometry, movement [24, 36], or task heuristics [26, 27], rather than a physiological proxy learned from measured muscle activity.

Physiological sensing offers a direct window into physical effort. Surface electromyography (EMG) provides a direct measure of muscle activation and has been widely used in ergonomics to quantify muscular load and fatigue. This paper investigates *NeckErgonomics*: an interaction approach that adapts HMD interfaces according to estimated neck muscle activity. The core concept is to use predicted neck muscle activity as a runtime interaction signal. *NeckErgonomics* estimates neck muscle activity associated with the user's current pose to decide when and how strongly the interface should adapt.

To make this practical without requiring live EMG sensing, we use an open-source EMG dataset [21, 22] to train a lightweight neural regression model that predicts relative neck muscle EMG data from head yaw, head pitch, and fixation duration. The resulting model provides an EMG-informed predictive signal using only standard HMD tracking.

We instantiate this concept in three applications, shown in Figure 1. First, in head-pointing, predicted neck EMG modulates the control-display gain, amplifying cursor movement when users adopt more physically demanding head poses. Second, in knowledge work, predicted neck EMG controls the recentering speed of a virtual window, using both head pose and fixation time to reduce sustained eccentric posture while preserving reading stability. Third, in 360° video viewing, predicted neck EMG modulates viewport amplification, allowing users to access surrounding content with less physical head rotation while limiting unnecessary camera movement. Together, these applications show how estimated neck muscle activity can be mapped to different interaction parameters: cursor gain, layout movement speed, and viewport rotation.

We evaluate the applications in a controlled within-subjects user study with 18 participants. The study is intended as a proof-of-concept evaluation of *NeckErgonomics* across three task contexts: active pointing, sustained knowledge work, and passive or semi-passive immersive media viewing. Across these tasks, the results show that model-based adaptation can improve neck ergonomics in different ways: by reducing accumulated head movement during pointing, reducing sustained head rotation during knowledge work, and reducing perceived physical demand and extreme head angles during 360° video viewing. These findings suggest that predicted neck muscle activity can serve as a reusable interaction signal across multiple HMD interaction scenarios. The study also reveals task-specific trade-offs with speed, precision, visual stability, and sickness, showing that adaptation should be tuned to the priorities of each interaction context. Our work conveys:

- *NeckErgonomics*, a model-informed approach for adapting HMD interaction according to predicted neck muscle activity.
- Three ergonomics-by-design applications for head pointing, knowledge work, and immersive media, showing how an EMG-grounded proxy can modulate cursor gain, window recentering, and viewport amplification.
- A proof-of-concept user study showing that these applications reduce head movement, sustained head rotation, and perceived physical demand across tasks, while revealing task-specific trade-offs.

By bridging predicted neck EMG with interaction-level adaptation, this work takes a step toward XR systems that are responsive not only to what users do but also to the physical cost the model predicts for doing so.

2 Related Work

2.1 Neck ergonomics in HMDs

Previous studies have investigated neck ergonomics when users use phones, laptops, and desktop computers [19, 25, 37, 38]. However, the relationship between neck ergonomics and HMD has received limited attention to date. HMDs have been shown to increase head movements (due to their reduced field of view) and muscle load (due to their additional weight) [1]. This has been observed in head pointing involving frequent movement [39] as well as in fixed head poses maintained over longer periods [11]. Other studies have reported neck muscle fatigue induced by head-worn equipment, such as helmets and night-vision goggles [6, 33]. Different head poses

and holding time would significantly affect neck muscle activity in HMD [20, 22]. Arm movement can also affect neck ergonomics, for example, during mid-air interactions in virtual reality (VR), leading to discomfort in the neck and shoulder regions [13]. These studies motivate interaction design aimed at improving neck ergonomics in HMD.

2.2 Models of neck muscle activity

Previous research has proposed several neck muscle activity prediction models. A previous study proposes a real-time neck kinematic model to simulate neck muscle activity. The model tracks head pose with an Inertial Measurement Unit and simulates neck muscle activity on the OpenSim platform¹ [15]. This work shows the value of modelling neck effort from head posture. However, for our purposes, the model is less directly suited to HMD interaction adaptation because it is not based on measured neck muscle activity (EMG) data and does not account for the additional weight of the HMD on the head.

Other prior work by Zhang et al. [39] studies and models neck muscle activity during pointing in HMDs. They collected sEMG data from electrodes placed on the splenius capitis (SPL) and sternocleidomastoid (SCM) muscles, which they used to train a deep learning model based on Convolutional Neural Networks (CNNs). This work is closely related to our motivation, as it connects head-based interaction with measured neck muscle activity. However, the model is mainly designed to estimate EMG after a movement has occurred, either based on a completed trajectory or a known target position, rather than to provide a continuous runtime signal for interface adaptation.

Together, these models show that neck muscle activity can be computationally estimated, but they are not directly designed for the interaction problem we address in this paper. Our model is grounded in physiological signals and predicts neck muscle activity from head yaw, pitch, and fixation time. It enables real-time adaptive techniques without requiring live EMG sensing.

2.3 Ergonomic adaptations in XR

Prior work has proposed several XR interaction adaptations to reduce neck demand, primarily by reducing head turning through UI relocation and rotation/gain remapping. However, these adaptations are typically driven by indirect cues, such as geometry, movement [24, 36], or task heuristics [26, 27].

Bring2Me detects that a widget/menu is outside the current field of view and brings it back into view for the user; its trigger is the visibility/FoV state [2]. Techniques for wide virtual workspaces reduce turning by manipulating the mapping between measured head orientation and display/workspace positioning. They use signals such as head yaw/pitch angles, crossing display boundaries, and head-ray transitions to apply counter-rotation, deadzones, or automatic transitions; the adaptation is thus keyed to geometric pose and layout events [24, 26]. In head-based pointing, dynamic control–display gain (e.g., HeadShift) adapts the mapping using head movement dynamics (e.g., speed/acceleration profiles) as a proxy to reduce head movement, again relying on kinematics rather

than muscle activity measures [36]. To address limited physical turning, a series of studies apply dynamic viewport rotation gains as a function of head rotation, using head orientation/angle thresholds as the primary signal [16, 29]. In contrast, recentring techniques gradually steer users back toward a forward-facing posture based on current head yaw offset and navigation/movement state [29, 32].

Together, these techniques demonstrate that adaptation can reduce the demand on the neck muscles, yet their triggers are predominantly geometry-, visibility-, or motion-based and cannot reliably indicate instant neck effort or physiological state. More importantly, these cues are often technique-specific: a field-of-view boundary, a head-speed profile, or a window position may work for one interface, but they do not provide a common signal of the user's bodily effort. Predicted neck muscle activity offers a more general and reliable adaptation signal because it reflects the physical demands that different techniques ultimately aim to reduce. This motivates *NeckErgonomics*: instead of only reducing movement through geometric or task-based rules, we estimate neck EMG associated with head poses and use that estimate to adapt interface behaviour.

3 Predicting Neck Muscle Activity

Motivated by the concept of adapting head-based interaction, we train a lightweight machine learning model on real EMG data to predict neck muscle activity without requiring sensing hardware.

To achieve this, we selected and preprocessed an open dataset and trained a lightweight neural regression model to predict users' neck muscle activity from head orientation and fixation time. Full preprocessing, model-comparison, and per-fold evaluation details are reported in the supplemental material.

Data and preprocessing. We trained on an open-source EMG dataset [21, 22], the most suitable to date for this purpose: 15 participants, six neck channels (extensor group, splenius capitis, sternocleidomastoid), spanning $\pm 80^\circ$ yaw and $\pm 50^\circ$ pitch with head-pose and fixation-time annotations. We applied per-participant standard scaling and an inverse transform to correct positive skew (see supplemental materials).

Architecture, training. We compared several regressors, including *MCLNet* [39], Random Forest, Gradient Boosting, and a Temporal Convolutional Network. We selected a feed-forward multi-layer perceptron (MLP) because it achieved the best R^2 at the lowest inference cost (see supplemental materials). The network has three input units (head yaw, head pitch, fixation time), two hidden layers (300 and 50 ReLU-activated neurons), and one output unit (estimated cumulative EMG); it was trained with Adam [14] (learning rate 0.001) and early stopping with a 10% validation split.

Model performance. We evaluated the model using participant-independent leave-one-participant-out cross-validation across the 15 participants. In each fold, the model was trained on data from 14 participants and tested on the held-out participant. The model achieved $R^2 = 0.718 \pm 0.110$, with low prediction error across folds (Table 1). The average inference time was 0.10 ms per prediction, making the model suitable for real-time XR adaptation.

Figure 2 shows the predicted EMG across head orientation and fixation time. Figure 2a and Figure 2b are low near the centre and

¹<https://opensim.stanford.edu/>, accessed: 2025-08-24

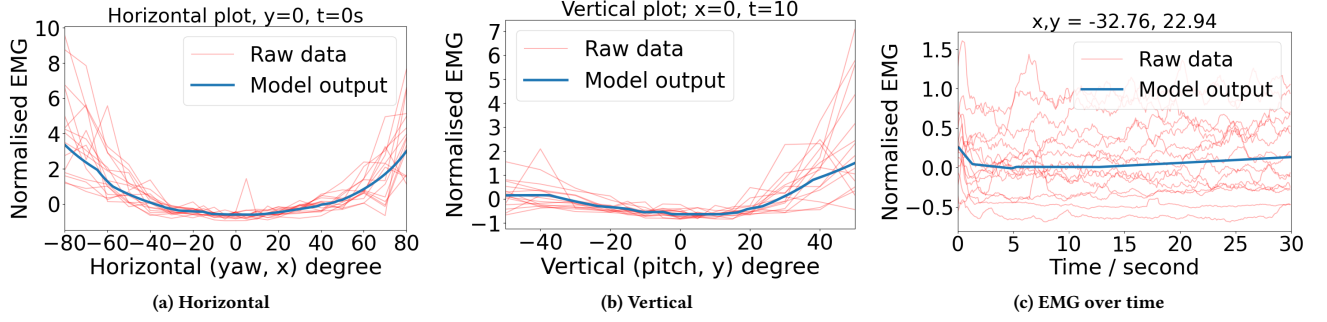


Figure 2: Model prediction examples for (a) horizontal head orientation, (b) vertical head orientation, and (c) EMG over time. The blue line is the model prediction; translucent lines in the background are scaled participant data. (a) Horizontal sweep with $x \in [-80^\circ, 80^\circ]$, $y = 0$ and $t = 0$. (b) Vertical sweep with $y \in [-50^\circ, 50^\circ]$, $x = 0$ and $t = 10$ s. (c) Temporal trace at the fixed pose $x = -32.76^\circ$, $y = 22.94^\circ$ from 0 to 30 s.

Table 1: The participant-independent model performance score for predicting neck muscle activity.

Metrics	Score
R^2	0.718 ± 0.110
Mean Absolute Error	0.026 ± 0.005
Mean Squared Error	0.001 ± 0.0003
Root Mean Squared Error	0.034 ± 0.005
Mean Absolute Percentage Error	0.095 ± 0.002
Normalised Root Mean-Square Error	0.114 ± 0.0179
Normalized Mean Absolute Error	0.087 ± 0.015
Runtime	0.10 ms

rise toward the periphery (bowl-shaped). Figure 2c decreases over roughly the first 3 seconds and then stabilises.

The model thus provides a real-time proxy for neck muscle activity that the applications in section 4 use to decide when and how strongly to adapt the interface, which in turn leads to different head movements or sustained eccentric poses. To prevent model output from having a negative value (for better scaling), we add 0.65 to all model output. After that, the minimum output is 0.

4 Applications: HMD Interaction Based on Predicted Neck EMG

In the following, we demonstrate how our neck muscle activity prediction model can be integrated into interactive systems to improve neck ergonomics in head-based interaction. We present three applications spanning head pointing, knowledge work, and immersive media, each exemplifying a different class of adaptation enabled by the same underlying model.

4.1 Head Pointing and Selection

Pointing and selection are basic operations in HCI. In a large-amplitude head-pointing task, a 1:1 control-display gain can lead to uncomfortable neck postures. On the other hand, a higher control-display gain could impair pointing accuracy.

In this section, we adapt the head cursor's gain using our model (see Figure 1, Left). Thus, we apply a higher gain the more the head rotates to reduce extreme eccentric angles, and a lower gain in the central comfort area to maintain precision. The amount of cursor amplification is defined as:

$$\text{Cursor}_{\text{Amplification}} = \text{model}(\text{yaw}_{\text{head}}, \text{pitch}_{\text{head}}, 0) \times k \quad (1)$$

with $\text{model}(\text{yaw}_{\text{head}}, \text{pitch}_{\text{head}}, 0)$ as the model output and k is a constant scaling the model output. Yaw_{head} and $\text{Pitch}_{\text{head}}$ are head yaw and pitch. Note that the model input "time" is always set to 0 in this technique because we treat head movement as not a fixation.

We use $\text{cursor}_{\text{amplification}}$ to adjust the cursor position as follows, with $\text{Yaw}_{\text{offset}}$ and $\text{Pitch}_{\text{offset}}$ being extra pitch and yaw degrees related to head orientation:

$$\begin{aligned} \text{Yaw}_{\text{offset}} &= \cos(\alpha) \times \text{cursor}_{\text{amplification}}, \\ \text{Pitch}_{\text{offset}} &= \sin(\alpha) \times \text{cursor}_{\text{amplification}}, \end{aligned} \quad (2)$$

with $\alpha = \text{atan2}(\text{pitch}_{\text{head}}, \text{yaw}_{\text{head}})$.

To limit cursor amplification and prevent the cursor from leaving the user's field of view, we clamp the value to 0 or the maximum amplification. A previous study on eye-head coordination in HMD shows a comfortable eye-in-head angle of around 15 to 25 degrees, depending on direction [30]. Thus, we set the maximum amplification in the upward direction to 15°, and in the other directions to 25°. With that, the $\text{Yaw}_{\text{offset}}$ and $\text{Pitch}_{\text{offset}}$ are never more than the maximum amplification in their specific direction. After a pretest, we set k to 30. This is a relatively high scaling factor so that the small model output in the near-central range still yields useful amplification (Figure 2). This high sensitivity to predicted muscle activity produces greater amplification, thereby reducing the required head movement, at the potential cost of pointing precision.

This application demonstrates how predicted neck muscle activity can serve as a continuous control signal for active head-based input. Head pointing requires frequent, moment-to-moment head movement, so the model is used here without fixation time to modulate control-display gain directly based on the current head pose and the predicted neck muscle activity. This illustrates the most immediate form of *NeckErgonomics*: when the head enters a pose associated with higher predicted neck muscle activity, the interface

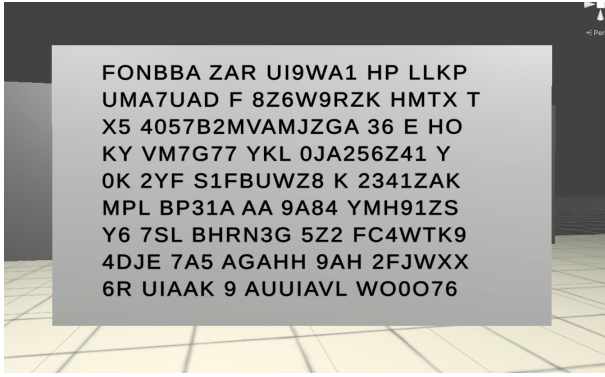


Figure 3: Knowledge work task: participant’s first-person view. Participants scanned a random string of capital letters, digits, and spaces and pressed a controller button for each occurrence of the target character “A”.

can amplify cursor movement to reduce further physical rotation; when the head remains near the central comfort area, the mapping stays closer to normal to preserve precision. The application, therefore, tests whether an EMG-informed proxy can replace purely kinematic gain heuristics in a basic interaction task.

4.2 Content Recentering During Knowledge Work

Working in VR, e.g., with multiple screens, promises to eliminate distractions and have a large area to place multiple windows [11]. However, having multiple windows necessitates placing them at angles that require large head movement [11]. Working (reading, watching, etc.) on such a large head angle requires maintaining high neck muscle activity, which could lead to neck fatigue. Gradually moving an active window to the front while the user looks at it can help prevent sustained eccentric rotations (see Figure 1, Centre).

In this application, our model adapts the window’s moving speed to the corresponding neck muscle activity, reducing neck angle while preserving the reading experience. In practice, this means the window would move faster at eccentric angles and slower as the head angle decreases. Specifically, when the window is at an eccentric angle, the model will modulate the window’s speed to accelerate recentering and bring the user’s head away from regions associated with higher predicted neck muscle activity. When the window is centred, the predicted EMG is low. Our model slows down recentering to preserve reading comfort, rather than forcing a 0° neck angle. Recentering speed v_{model} is defined as follows:

$$v_{model} = model(yaw_{head}, pitch_{head}, t) \times k + 1 \quad (3)$$

$model(yaw_{head}, pitch_{head}, t)$ is the model output and k is a constant scaling the model output. Yaw_{head} , $pitch_{head}$ are head yaw and pitch. t is the window active time, used here as an approximation of fixation duration on the active window.

Previous studies show that the comfortable gaze smooth-pursuit speed during reading is optimal below 7°/s with an upper practical limit of about 14°/s for sustained comprehension [34]. To limit visual discomfort while reducing sustained eccentric head rotation, we

chose to clamp the v_{model} between 1 and 9°/s based on informal pretests. As the maximum output of the model (horizontal) is approx. 5 (cf. Figure 2a), we set k to 1.6 to keep v_{model} below 9.

This application shows that *NeckErgonomics* is not limited to remapping input movement. In knowledge work, the main ergonomic issue is often not a single large head movement, but sustained attention to content placed at an eccentric angle. This makes fixation time important: a brief glance toward an off-centre window may be acceptable, whereas holding the same head pose while reading can gradually increase physical demand.

This application uses the model’s fixation-time input to make adaptation sensitive to both the head’s location and the fixation duration. The model controls layout adaptation by determining how quickly the active window should move back toward the user’s forward direction. Rather than forcing all content to the centre immediately, this demonstrates a time-aware form of ergonomic assistance: the system can intervene more strongly when eccentric posture is sustained, while preserving visual stability and reading comfort when the demand is lower.

4.3 360° Video Watching

360° videos are popular forms of VR entertainment. However, full exploration requires excessive neck rotation as the content may be displayed around the viewer. While players use alternative viewport control techniques to reduce head movement and expand viewing range [17, 18, 32], their development is complex [18], induces motion sickness [23], or requires manual input (e.g., Controller Drag on YoutubeVR).

In this section, we leverage our model to amplify the viewport to improve the 360°-video experience. Specifically, the model increases viewport rotation to reduce neck angle, even at the risk of inducing motion sickness at eccentric head rotation angles, while in central regions it reduces amplification to preserve viewing comfort (see Figure 1, Right). Our model modulates viewport rotation as follows:

$$Yaw_{viewport} = yaw_h + model(yaw_h, pitch_h, 0) \times k \quad (4)$$

$model(yaw_h, pitch_h, 0)$ is the model output and k is a constant scaling the model output. Yaw_h and $pitch_h$ are head yaw and pitch. As in head pointing (subsection 4.1), we treat head movement as non-fixation and set the model’s time input to 0. We only amplify the horizontal viewport [18, 29, 35]. Note that this differs from the pointing application (cf. subsection 4.1), where we modified both pitch and yaw.

We configured the system so that users can look 180° behind them when their heads rotate 90° (maximum model output is approx. 5, $k = 15$, cf. Figure 2a) While a higher k reduces total neck rotation, it also increases the risk of cybersickness due to fast camera movement [29, 35]. The model gain was clamped between 1 and 2, ensuring that a 180° viewing angle on one side is achievable with a 90° head rotation.

This application shows how the same model can support passive or semi-passive viewing, where users are not selecting targets or reading fixed content but exploring a surrounding scene. In this case, predicted neck muscle activity modulates viewport amplification, allowing the system to provide more assistance when the model predicts higher neck muscle activity and less assistance near the central viewing region.

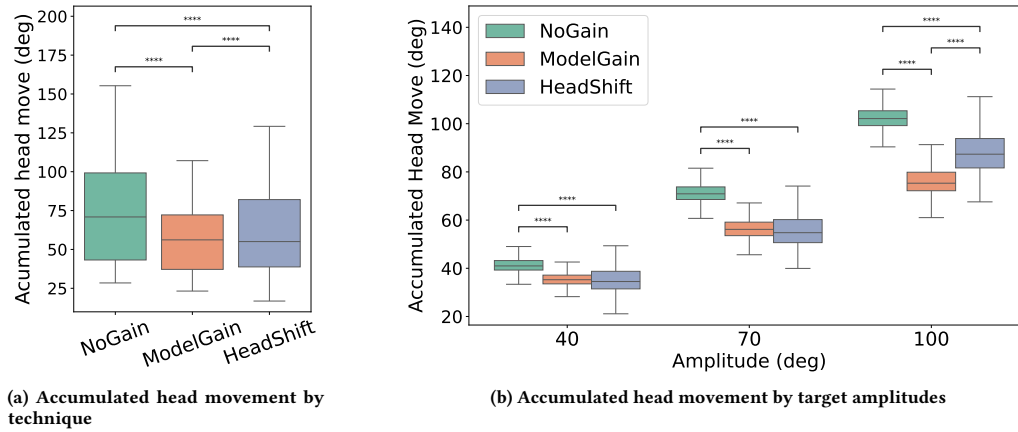


Figure 4: Accumulated head movement during head-pointing: (a) by technique, (b) by target amplitude. ModelGain reduces head movement compared to NoGain and HeadShift, with the gap widening at 100° amplitude.

This illustrates a third class of *NeckErgonomics* adaptation: changing the relationship between physical head movement and virtual camera movement. The application exposes a different trade-off from pointing or window recentering: reducing neck rotation must be balanced against perceptual comfort and potential sickness from camera amplification.

5 Evaluation of NeckErgonomics

Having defined the three applications in section 4, we evaluate them in one within-subjects study with $N = 18$ participants. Each participant completed the three tasks described below.

We evaluated each application against two baselines, a no-adaptation baseline and an alternative adaptation technique:

- **Head pointing** (Fitts-Law-like task): **ModelGain** vs. **NoGain** (1:1 mapping) and **HeadShift** [36].
- **Knowledge work** (reading task): **ModelSpeed** vs. **NoSpeed** (no recentering) and **ConstantSpeed** (3°/s, the midpoint of the comfortable smooth-pursuit reading range [34]).
- **360° video** (seated viewing): **ModelGain** vs. **NoGain** (1:1 mapping) and **HeadGain** (1:2 amplification) [28].

5.1 Tasks

Participants completed three tasks, each evaluating one application against two baselines.

5.1.1 Head pointing and selection. We compare the techniques in a Fitts-Law-like task. We modified the standard task to have larger target amplitudes beyond the field of view ($>40^\circ$) [5] to better reflect the large head rotation. We asked participants to point at targets with their heads and confirm their selection with the controller’s trigger button. We had 11 targets arranged in a circle [36]. Following a sequence, participants must select 23 times to complete a round. We discarded the first trial in each round. In the task, we varied the amplitude (40°, 70°, and 100°), the target size (2° and 4° diameter), and the techniques (ModelGain, HeadShift, and NoGain).

5.1.2 Content recentering during knowledge work. To evaluate recentering, we asked participants to do a reading task. The task is inspired by a previous study on evaluating the image quality of visual displays [3]. Participants saw a text displayed on a white window, positioned at yaw angles of $\pm 40^\circ$ and $\pm 80^\circ$ in the horizontal plane (Figure 1). The text consists of random strings of capital letters, digits, and spaces, with 9 lines and 25 characters per line, containing 18–22 “A” (Figure 3). We instructed participants to scan for and locate the target character “A” and press a controller button for each occurrence. Once they were done, they pressed another button to switch to the next trial. Each trial took about 20 seconds. Once the participants’ head gaze reaches the window, which appears at a starting position of $\pm 40^\circ$ or $\pm 80^\circ$, recentering begins. Participants would do 4 trials for each position, with 2 on the left and 2 on the right.

5.1.3 360° video watching. Participants were asked to watch a 360° video (*HELP*, 2016²) for around 4 minutes with ModelGain, HeadGain, and NoGain. Each participant watched the video three times, once per technique.

5.2 Procedure

Upon arrival, we briefed participants on the experiment, and they completed consent forms and a demographic questionnaire. Participants were seated and fitted with the HMD. After that, the participant experienced the three tasks defined in subsection 5.1: pointing and selection, knowledge work, and 360° video watching. The order of tasks was random. The sequence of techniques was counterbalanced, ensuring that each technique appeared equally often in 1st/2nd/3rd place. Before starting, we explained the task, and the participants practised until they were comfortable. After each task, they removed the headset to complete questionnaires. Breaks were offered between tasks. The study was approved by the institutional ethics board, took around 90 minutes, and participants were compensated with 15 GBP. During the study, participants

²<https://youtu.be/G-XZhKqQAHU?si=1c2t4QH6I-hKjCa>, 2025-05-15

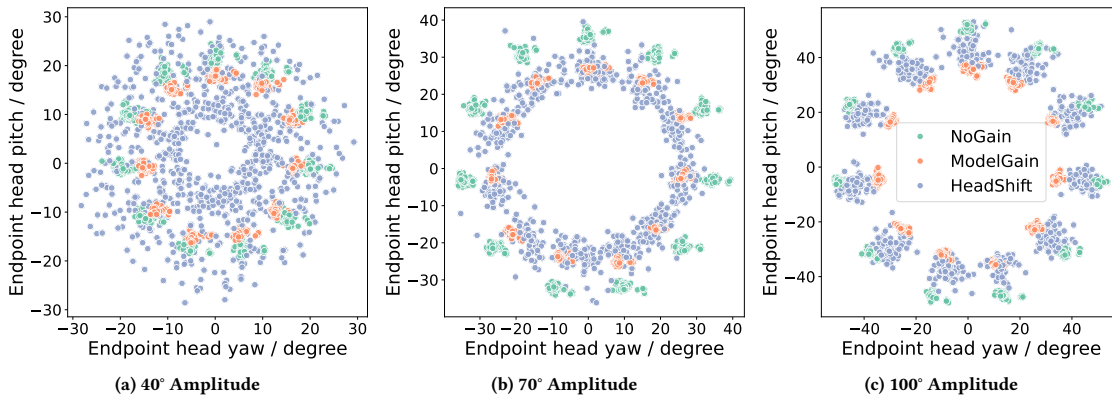


Figure 5: End Point Distribution for 40°, 70° and 100° Amplitudes

were required to sit on a stationary chair and keep their lower limbs still. A brief, unstructured interview would be conducted at the end of each task, focusing mainly on the user's comments on the technique and the task.

5.3 Participants

We recruited 18 participants (11 female, 7 male) from a local university, with an average age of 29.4 years ($SD = 8.6$, range 22–53). None reported neck, shoulder, or dermatological issues. Around 88% had limited VR experience: 3 had none, 12 used it rarely, and 1 used it monthly.

5.4 Apparatus

We implemented the scenarios and the individual techniques with Unity 2022.3.49f1. We used a Meta Quest 3 VR headset for the study, with a 146° diagonal rendered FOV, 2064×2208 per-eye resolution, and a 72 Hz refresh rate on a computer with an Intel Core i7-12700 CPU, 16 GB of RAM, and an NVIDIA GeForce RTX 3070 Ti GPU.

5.5 Measures

Across all three tasks, we collected the RAW NASA-TLX [7] and 7-point Likert ratings of preference and comfort after each technique. We asked “How comfortable is the technique?” (1 = “not at all”, 7 = “very comfortable”) and “How much do you like the technique?” (1 = “not at all”, 7 = “very much like it”). Head pose was recorded throughout; the head angle was measured relative to the world by the headset. For the knowledge-work and 360° video tasks, participants additionally completed the Virtual Reality Sickness Questionnaire (VRSQ) [12]. Per-task measures were:

- **Head pointing.** Completion time, cumulative head movement (°), throughput (ISO 9241-411 [5]), endpoint distribution, and error rate.
- **Content recentering.** Count of target character “A” and completion time per trial; error count is the difference between total and confirmed “A”s. Head position is also used to derive head movement and head-angle distribution.
- **360° video.** Per-frame head angle and virtual camera angle.

Note, we did not record neck EMG, muscle fatigue, or musculoskeletal load during the study. Our measurements, cumulative head movement, sustained head rotation, and perceived physical demand, are therefore only indirect proxies for neck ergonomics.

We use repeated-measures ANOVAs for objective measures ($\alpha = .05$). We apply Greenhouse–Geisser corrections when Mauchly's test indicates a violation of sphericity, and the Aligned Rank Transform when the Shapiro–Wilk test indicated non-normality. Post-hoc tests are Bonferroni-corrected. Subjective measures (Raw NASA-TLX, comfort, preference, VRSQ) were tested with Wilcoxon signed-rank tests. Effect sizes are reported as η^2 for ANOVAs and rank-biserial r for Wilcoxon tests. We discarded 36 pointing trials (0.51%) whose completion time exceeded three standard deviations from the per-condition mean.

5.6 Results

Full statistical detail and per-technique subjective-metric distributions are reported in the supplemental material.

5.6.1 Head pointing and selection.

Accumulated head movement. ModelGain reduced total head movement compared to baselines (Technique main effect, $F(2, 289) = 445$, $p < .001$, $\eta^2 = 0.75$; post-hoc NoGain > HeadShift > ModelGain, all $p < .001$). A Technique $\times$ Amplitude interaction showed the gap widening with amplitude ($F(2.33, 39.6) = 106.7$, $p < .001$, $\eta^2 = 0.83$); at 100°, ModelGain further reduced head movement compared to HeadShift ($p < .001$, see Figure 4 and Figure 5).

Trade-offs: completion time, error rate, throughput. Completion time was longer with ModelGain than with both baselines (Technique main effect, $F(2, 34) = 24$, $p < .001$, $\eta^2 = 0.13$; post-hoc ModelGain > NoGain and ModelGain > HeadShift, both $p < .001$). Error rate showed a Technique main effect ($F(2, 289) = 16$, $p < .001$, $\eta^2 = 0.10$): HeadShift had a lower error rate than ModelGain ($p < .001$), driven by the 2° target size; ModelGain and NoGain did not differ. Throughput followed the same trade-off (Technique main effect, $F(2, 34) = 48$, $p < .001$, $\eta^2 = 0.32$): NoGain > ModelGain > HeadShift (all $p < .001$).

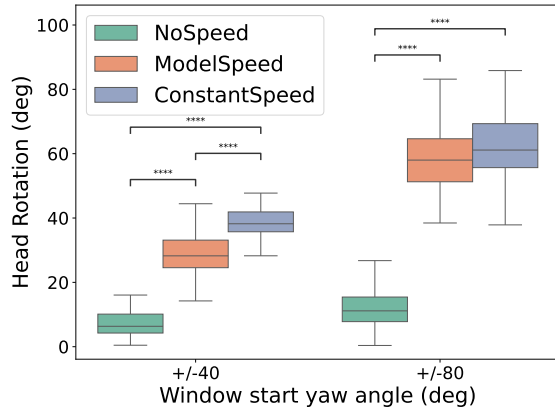


Figure 6: Head rotation during the knowledge-work task, by technique and window start angle. ModelSpeed and ConstantSpeed reduce head rotation compared to NoSpeed at both starting positions; ModelSpeed beats ConstantSpeed at $\pm 40^\circ$.

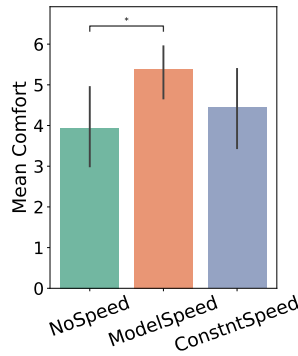
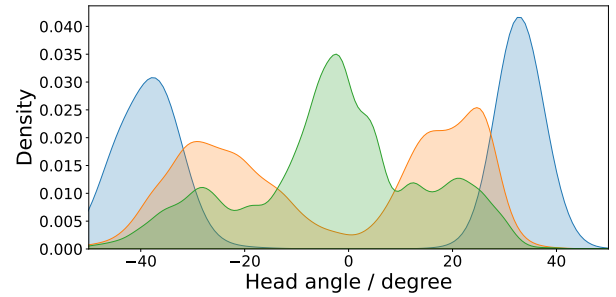


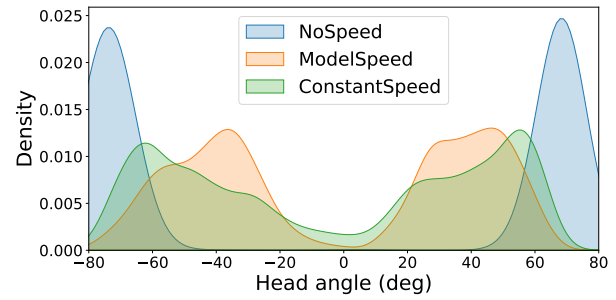
Figure 7: Mean comfort by technique during the knowledge-work task (7-point Likert scale, 95% CI). NoSpeed is rated significantly less comfortable than ModelSpeed.

Endpoint Distribution. Figure 5 shows the endpoint distribution of all three techniques in all amplitudes. The endpoint is the head pose (pitch, yaw) when pressing the selection button in a trial. Endpoints under ModelGain were tightly clustered. The Hopkins clustering statistic [8] was 0.960 (ModelGain) and 0.965 (NoGain) versus 0.803 (HeadShift), and the standard deviation of accumulated head movement was 4.91 (ModelGain) and 5.50 (NoGain) versus 8.35 (HeadShift). This is consistent with HeadShift's relative mapping, which is known to suffer from cursor drift (an unexpected offset between cursor and head gaze).

Subjective metrics. We found no significant differences in Raw NASA-TLX, preference, or comfort across techniques (all $p > 0.10$). The pointing results show that ModelGain substantially reduced accumulated head movement compared with both NoGain and HeadShift, with the strongest benefit at the largest target amplitude. This supports the value of using predicted neck muscle activity as



(a) Head angle distribution in $\pm 40^\circ$ window starting angle



(b) Head angle distribution in $\pm 80^\circ$ window starting angle

Figure 8: Head angle distribution by techniques and window starting angle with all participants' data.

a runtime signal to reduce the required head movement in large-amplitude head pointing.

The benefit came with a performance trade-off: ModelGain increased completion time, and HeadShift achieved lower error rates for small targets. This suggests that EMG-model-informed gain introduces a distinct optimisation objective for head-based pointing. Rather than optimising only for speed and precision, ModelGain incorporates predicted neck muscle activity as an additional control signal during interaction, especially when large or repeated head movements are required.

5.6.2 Content recentering during knowledge work.

Head rotation. Both adaptive recentering techniques reduced head rotation compared to the fixed-window baseline (Technique main effect, $F(1.42, 24) = 406$, $p < .001$, $\eta^2 = 0.89$; post-hoc NoSpeed > ConstantSpeed > ModelSpeed, all $p < .001$, see Figure 6). A Technique $\times$ window-start interaction ($F(2, 34) = 62.3$, $p < .001$, $\eta^2 = 0.41$) showed that at $\pm 40^\circ$ window start, ModelSpeed reduced head rotation more than ConstantSpeed ($p < .001$), while at $\pm 80^\circ$ the two adaptive techniques converged. As expected, $\pm 80^\circ$ starts produced more head movement than $\pm 40^\circ$ ($F(1, 17) = 272$, $p < .001$, $\eta^2 = 0.72$).

Completion time and error. ConstantSpeed had a shorter completion time than ModelSpeed ($F(2, 85) = 6.8$, $p = 0.0017$, $\eta^2 = 0.14$; post-hoc $p = 0.0012$). Error rate did not differ across techniques (grand mean 0.090, 95% CI [0.081, 0.097]; all $p \geq .275$).

Comfort. Comfort differed across techniques ($p = 0.033$, $r = 0.19$, see Figure 7): NoSpeed (mean 3.94, 95% CI 1.05) was rated less comfortable than ModelSpeed (mean 5.38, 95% CI 0.73, post-hoc $p = 0.018$). VRSQ and Raw NASA-TLX showed no significant differences (all $p \geq .272$).

Reading behaviour. Figure 8a and Figure 8b show participants' head angle distribution in starting angles. Figure 8a demonstrates the majority of ConstantSpeed distributed in the centre area, while ModelSpeed is mainly distributed in $\pm 20^\circ$ to $\pm 30^\circ$. This implies that the ConstantSpeed is faster than the ModelSpeed for this starting angle. Figure 8b demonstrates the majority of ConstantSpeed distributed in $\pm 70^\circ$ to $\pm 50^\circ$, while ModelSpeed is mainly distributed in $\pm 50^\circ$ to $\pm 30^\circ$. This implies that the ModelSpeed is faster than the ConstantSpeed at the beginning of the task, but slows down as the head angle decreases. NoSpeed head angle is always close to the starting angle.

The recentering results show that both recentering techniques reduced head rotation relative to the fixed-window baseline, with ModelSpeed producing the lowest overall head rotation. This supports the value of using predicted neck muscle activity and fixation time to reduce sustained eccentric posture during reading-like work. However, the results also show that stronger recentering is not always better. ConstantSpeed reduced head rotation but resulted in lower comfort and greater instability for some users, whereas ModelSpeed better balanced recentering with reading stability. This suggests that model-based recentering is useful not because it re-centers as fast as possible, but rather because it provides a signal for deciding when stronger assistance is warranted.

5.6.3 360° video watching.

Physical demand and sickness. On the NASA-TLX, physical demand differed across techniques ($p = 0.008$, $r = 0.27$): participants reported higher physical demand with NoGain than with ModelGain ($p = 0.034$). Other NASA-TLX subscales did not differ (all $p \geq .152$). On the VRSQ (Figure 9), total sickness score differed across techniques ($p = 0.033$, $r = 0.19$): HeadGain induced more motion-sickness symptoms than NoGain ($p = 0.036$). Oculomotor and disorientation subscales did not differ (all $p \geq .085$). Together, ModelGain reduced physical demand without HeadGain's increase in sickness symptoms.

Viewing behaviour. ModelGain and HeadGain reduced extreme head angles compared to NoGain: head angle distributions decayed to zero around $\pm 100^\circ$ under ModelGain and HeadGain but extended out to $\pm 170^\circ$ under NoGain. ModelGain occupied a broader central head-angle band (0° to $\sim 50^\circ$) than HeadGain (0° to $\sim 35^\circ$), with NoGain spanning the widest range (0° to $\sim 80^\circ$). Despite the smaller head excursions, viewport coverage reached the full 360° under both ModelGain and HeadGain, while NoGain covered only $\sim 300^\circ$.

The 360° video results show that ModelGain reduced perceived physical demand compared with NoGain, while avoiding the increase in sickness observed with HeadGain. Viewing behaviour further suggests that ModelGain reduced extreme head angles while preserving broad viewport coverage. This result highlights why constant amplification is insufficient for immersive media. Reducing physical head rotation through camera amplification can introduce visual-vestibular discomfort, especially when high gain is applied

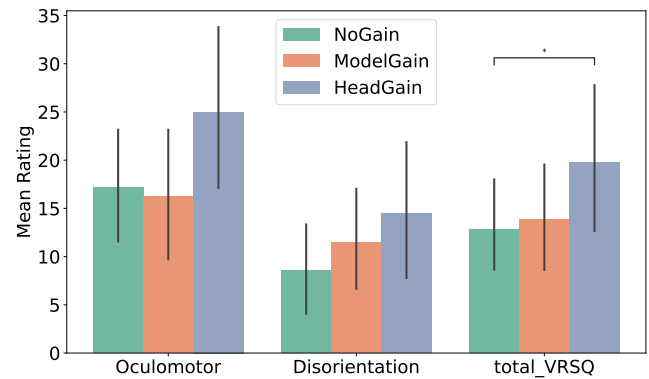


Figure 9: Mean Virtual Reality Sickness Questionnaire (VRSQ) scores during 360° video watching, with 95% confidence intervals. HeadGain induces significantly more motion-sickness symptoms than NoGain; ModelGain stays close to NoGain.

across the field of view. ModelGain offers a more selective alternative: it increases assistance when predicted neck muscle activity is higher, while keeping the mapping closer to normal near the central viewing region.

6 Discussion

6.1 Ergonomics-by-design by Adding an Ergonomic Cost

The proof-of-concept points to the value of *ergonomics-by-design* interaction techniques: ergonomics is an objective from the start of design, rather than a property assessed after the technique has been built. Ergonomics-by-design embeds an ergonomic cost into the system's behaviour: with *NeckErgonomics*, the interface adjusts assistance based on the estimated muscle activity of the current head pose. In our applications, interface behaviour is adapted not only to complete the task, but to redistribute head movement away from demanding poses. The three applications instantiate this principle through cursor gain, window movement, and viewport amplification. This opens a design space for HMD techniques that balance task performance with the physical cost of head movement, especially during longer sessions, in productivity scenarios, in immersive media, and for users with different physical capacities.

6.2 One signal for multiple interaction techniques

A key advantage of *NeckErgonomics* is that the same predicted neck muscle activity signal can support multiple interaction techniques. Prior ergonomic adaptations in XR often rely on task-specific rules: a window is recentred when it falls outside a comfortable viewing region, or a cursor gain is adjusted when the head is moving quickly. These rules can work well, but they are usually tied to a particular technique. In contrast, predicted neck muscle activity provides a reusable signal. In our applications, the same model output is mapped to three different parameters: cursor gain, window recentering speed, and viewport amplification. Each mapping still

needs task-specific tuning, but the underlying ergonomic logic is shared: when the predicted neck muscle activity of the current head pose increases, the interface can provide stronger assistance. This makes *NeckErgonomics* useful not only as a set of applications, but as a general design approach for creating EMG-model-informed XR interactions.

6.3 Ergonomics as a design trade-off

Across the three tasks, EMG-model-informed adaptation comes at a cost: each technique reduces head movement or sustained head rotation by incurring a cost elsewhere, and the balance differed by task. In head-pointing, ModelGain produced the largest reduction in accumulated head movement among the techniques tested, but paid for it with longer completion times and lower accuracy on the smallest targets. In knowledge work, ModelSpeed achieved a low sustained head rotation and improved comfort while preserving reading accuracy, at the cost of slower completion than the fastest fixed-speed baseline. In 360° video, the trade was more favourable: ModelGain lowered perceived physical demand and kept sickness at no-gain levels, avoiding the rise in symptoms caused by constant-gain amplification.

The value of the model-informed signal is that it allows designers to adjust trade-offs. The scale parameter k makes this choice explicit: it sets how strongly a technique intervenes, and therefore where it sits between reduced head movement and the competing demands of speed, precision, visual stability, and sickness. The appropriate setting is task-dependent and should follow each application's priority rather than a single global value.

For example, we set each k by first considering the fundamental limits from human-factors evidence, such as the comfortable eye-in-head range for pointing (15–25°) [30], comfortable smooth-pursuit speed for recentering (below 9°/s [34]), and the viewport geometry range for video. We then choose k so that the model's output range (from near 0 at the centre to about 5 at the periphery) spans that limit in recentering and video.

6.4 Future Work and Limitations

Despite the results of our work, it does not come without limitations, offering opportunities for future work. Our model does not yet accurately represent the variability between participants. Thus, it outputs the same predictions for all users, regardless of differences in neck muscle strength, mobility, or comfort. We also did not measure live EMG during interaction; predictions are computed from head pose and fixation time using a model trained on offline EMG data, so the proxy used in the applications is physiologically grounded but not a per-user measurement. Additional verification with direct measurement is needed. Relatedly, the model was trained on a dataset that included only a few young, healthy adults, which prevents generalisation to other user groups. Similarly, the sample of our user study consisted only of young healthy adults. Thus, generalisation to other user groups, e.g., with limited mobility, people with disabilities, or older adults, is necessary to verify the applicability of our work across the general population. Beyond participant demographics, several conditions limit this proof-of-concept's external validity. Each task used a single stimulus: one pointing layout, one reading text, and one 360° video. Because each

technique only helps when the head moves or sustains an eccentric pose, the size of its benefit depends on how much head movement a particular stimulus demands. Our effects are thus tied to these specific stimuli. Tests with more complex scenarios are necessary to further test *NeckErgonomics*. In addition, our study was seated and lab-based. Realistic productivity or media use, and standing or room-scale settings, may impose different neck loads. Also, the model was trained on one headset (HTC Vive Pro Eye) and deployed on another (Meta Quest 3), without being validated against live EMG data from our participants, so its in-context accuracy is unverified across devices and users. Finally, our study demonstrates short-term reductions in head rotation across three tasks, but it does not establish long-term reductions in fatigue, discomfort, or musculoskeletal risk. Longer-session studies, more diverse cohorts, including users with limited mobility and older adults, are necessary to determine whether the per-trial reductions observed here translate into sustained ergonomic benefit.

7 Conclusion

This work introduced *NeckErgonomics*, an interaction approach that adapts HMD interfaces according to estimated neck muscle activity. Using an open EMG dataset, we trained a lightweight model that estimates neck muscle activity from head yaw, pitch, and fixation time. This makes physiological grounding available through standard HMD tracking.

We integrated this proxy into three applications: modulated cursor gain for head pointing, dynamic window recentering for knowledge work, and adaptive viewport amplification for 360° video viewing. Our proof-of-concept study showed that these techniques reduce head movement, sustained head rotation, and perceived physical demand across different task settings, while also revealing trade-offs with performance, reading stability, and sickness.

Together, these results show that neck ergonomics can become part of the interaction logic itself, rather than only an outcome measured after design. By connecting predicted neck muscle activity with interface adaptation, *NeckErgonomics* points toward HMD systems that are responsive to physiological state.

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